Intro to AI in Medicine

Lec 13: Fairness & Bias in Medical AI

Last Time

- Project Intro
- Read over PowerPoint, word doc is streamlined version
- I will be available for questions after class

This Time

- Exploring the ways AI algorithms exhibit bias, harmful and otherwise
- Recap of bias from other lecture
- Let what you've learned about AI algorithms inform your opinions
- Lots of discussion! You guys know more than me
- Topics can be heavy so let's tackle them with an open mind

Bias in Data

- As we attempt to extract patterns from our data we must beware of sources of bias

- Bias really falls in one of two buckets:
  - Our data does contain, but is due to another factor not captured in data
  - Our data fails to capture

- How we formulate our problem has a massive impact!
  - What are we hoping to extract from the data
  - How do we preprocess data to exclude or counterbalance sources of bias

Types of bias we discussed…

- Historical Bias
  - Patterns in data that reflect inequalities that have occurred as a result of some factor not captured in the model
  - Ex. Amazon hiring
- Sample Bias
  - Sample not representative of whole population
  - Ex. Underinsured patient population not present in rich private hospital
- Aggregation Bias
  - Collapsing together clinically meaningful but different subgroups
  - Thymus read as pathology on pediatric chest x-rays
- And many more…

Recall

- Biased data leads to biased outcomes
    - Ex. Regression model – any inequities will be learned and propagated
    - Data cleaning involves removing these markers from data
- The decision you want to inform should be in mind when designing model
- Loss function is critical – what is your outcome metric
    - Minimizing readmissions will lead to different decisions than minimizing expenditures

Case Study

Artificial intelligence predicts patients’ race from their medical images

Study shows AI can identify self-reported race from medical images that contain no indications of race detectable by human experts.

- How does this work?
  - Input data
  - Problem type
  - Training regime

Case Study

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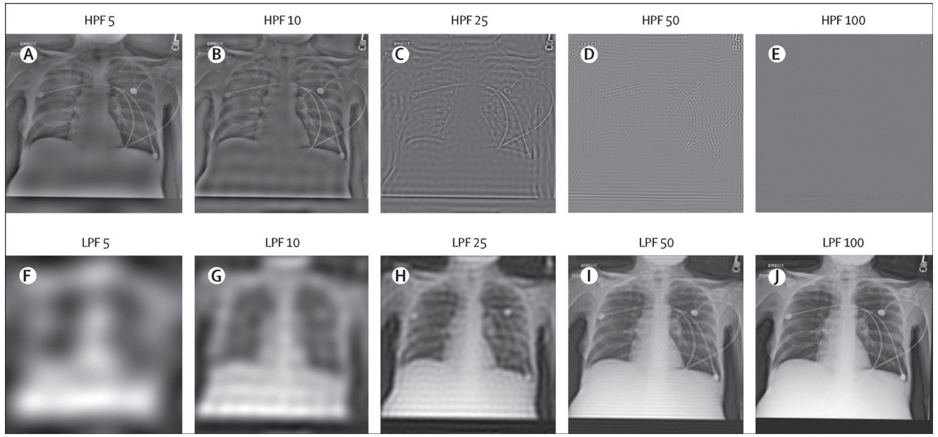
- Results:
  - AUROC = .96-.94 for prediction of race, even when filtered to only show borders of body structures (done to eliminate bone density)
  - When only using tabular data, AUROC fell to 0.54-0.61
- What implications does this have for clinical practice?
- What does this say about what models are learning?



Case Study

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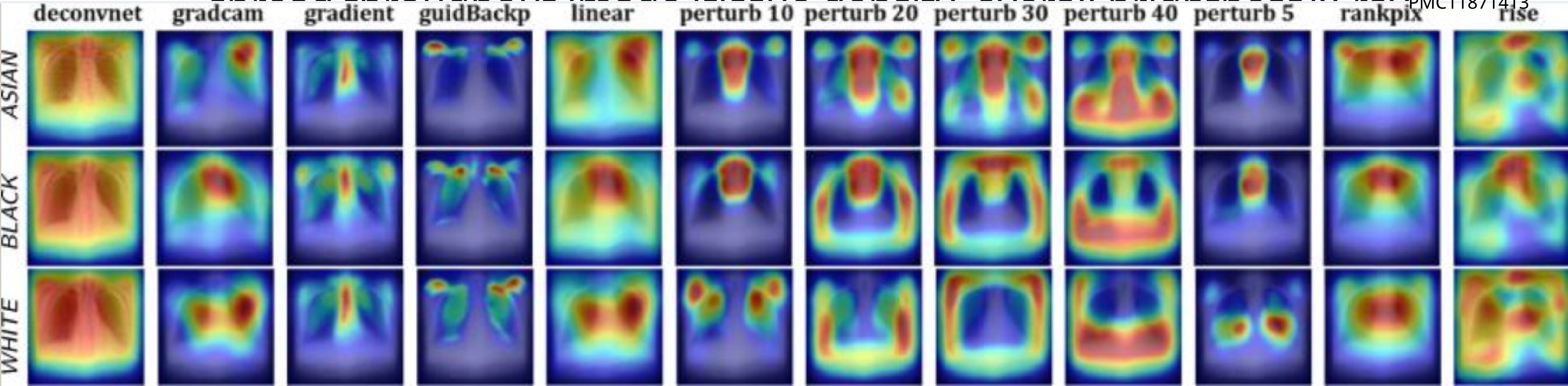
- How can we learn more about what these models are using to predict?
  - GradCam - > backpropagate which features have large role
  - Perturb - > find regions, which when removed, change prediction most
  - Low frequency features: those features which are subtle and spread throughout image (tissue density, overall brightness/x-ray

https://pmc.ncbi.nlm.nih.gov/articles/PMC11871413



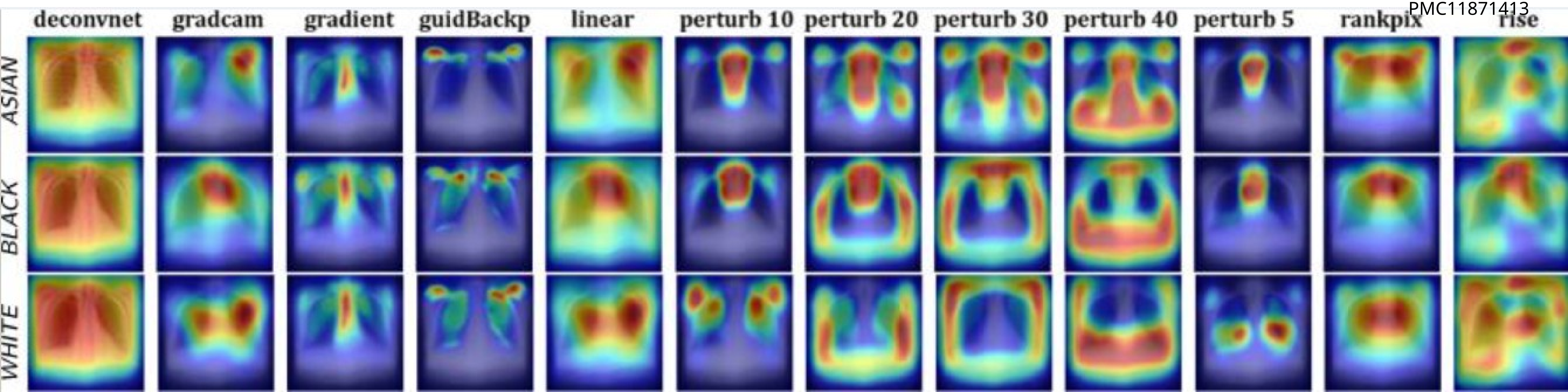
Case Study

Artificial intelligence predicts patients' race from their medical images

Study shows AI can identify self-reported race from medical images that contain no indications of race detectable by human experts.

- How can we learn more about what these models are using to predict?
  - Unintended proxies - > certain scanner scans more people of one race, has certain metadata, can be used to predict race
  - Better logging can help us identify and remove (or at least class balance) these features.

https://pmc.ncbi.nlm.nih.gov/articles/PMC11871413



Case Study

- Should we fight this?
  - Is it necessary to remove this information?
  - What level of accuracy is worth sacrificing for anonymization?
  - How can we evaluate if the identification of race/other features is causing our model to make predictions which have a clinical impact?
- If so, how?
  - Learn what features are important, then remove them
  - Train models to optimize performance on the *worst-performing subgroup*
  - Adversarial debiasing
  - Balance data beforehand

Definitions of Fairness

- Demographic Parity (Statistical Parity)
  - The AI system produces the same proportion of positive predictions across different groups
  - Example: A diagnostic tool flags 15% of patients for follow-up testing regardless of race/gender
  - *Medical consideration*: This may not be clinically appropriate if disease prevalence actually differs between groups
- Equalized Odds
  - True positive rates (sensitivity) AND false positive rates are equal across groups
- Equal Opportunity (similar to above)
  - True positive rates (sensitivity) are equal across groups

Definitions of Fairness

- Predictive Parity (Positive Predictive Value)
  - when the model says "positive," it's equally likely to be correct across groups
  - Example: When an AI flags a suspicious lesion, it's actually cancer 70% of the time for all groups
  - *Medical consideration*: Important for resource allocation decisions

The Impossibility Theorem

- **You generally cannot satisfy all fairness metrics simultaneously** when base rates (disease prevalence) differ between groups.
- This is known as the "fairness impossibility theorem.“
- For example, if Disease X affects 20% of Group A but only 5% of Group B:
  - Maintaining equal false positive rates AND equal true positive rates will violate demographic parity
  - Achieving demographic parity will likely harm calibration

Practical Steps to Mitigate Bias

- **Planning Phase**
  - Define fairness goals explicitly before model development
  - Identify relevant protected attributes (race, gender, age, socioeconomic status)
  - Engage stakeholders from affected communities
  - Determine which fairness metrics are appropriate for your clinical use case
- **Data Collection & Preparation**
  - Ensure diverse, representative training data across all relevant subgroups
  - Document data provenance and collection methods
  - Examine and report demographic composition of datasets
  - Check for and address selection bias in recruitment
  - Validate that labels are consistent and unbiased across groups

Practical Steps to Mitigate Bias

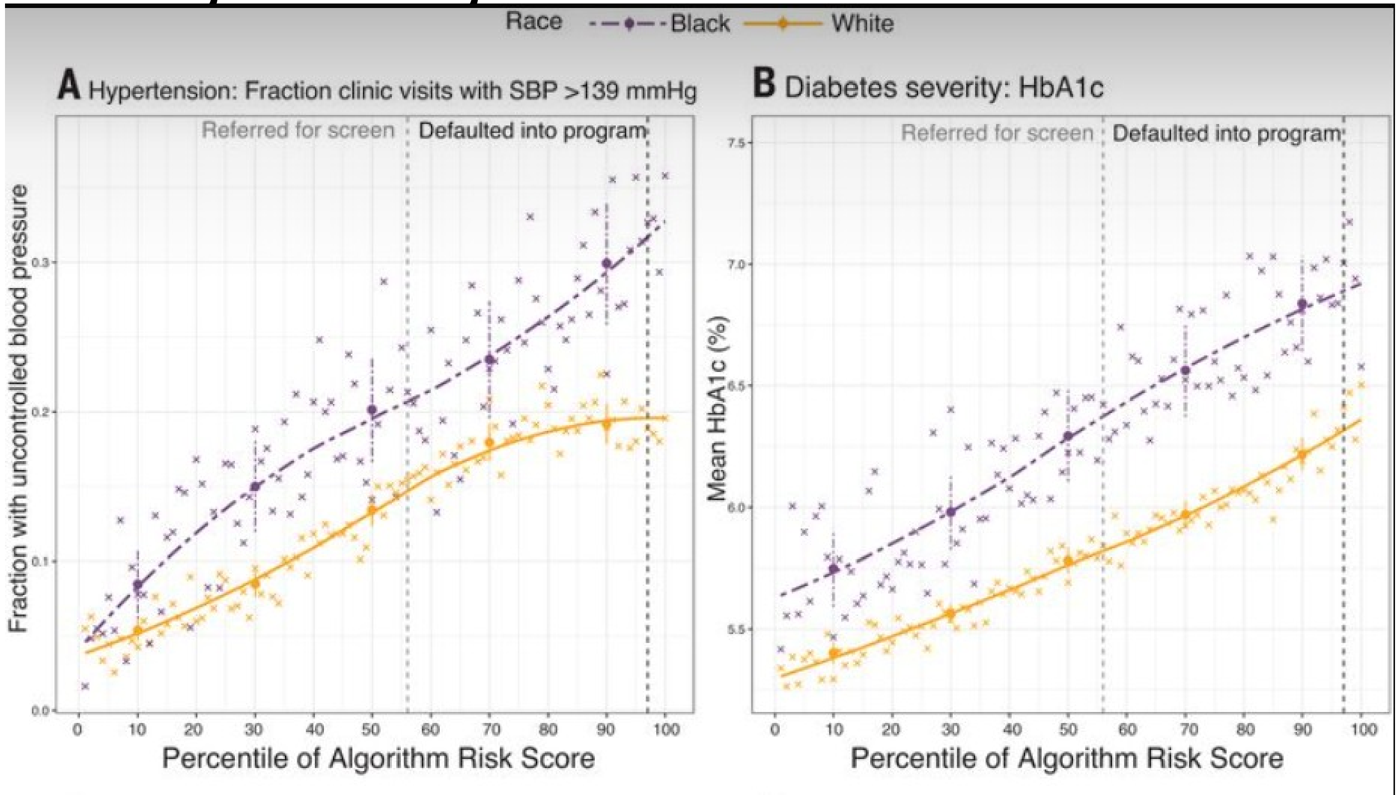
- **Model Development**
  - Stratify data analysis by demographic subgroups from the start
  - Avoid using race/ethnicity as direct features unless clinically justified
  - Be cautious of proxy variables that correlate with protected attributes
  - Consider fairness-aware training methods (adversarial debiasing, constrained optimization)
  - Use interpretable models when possible to identify potential sources of bias
- **Evaluation & Validation**
  - Report performance metrics separately for each demographic subgroup
  - Evaluate multiple fairness metrics (not just overall accuracy)
  - Test on external, diverse validation sets
  - Assess whether the model uses demographic "shortcuts" inappropriately

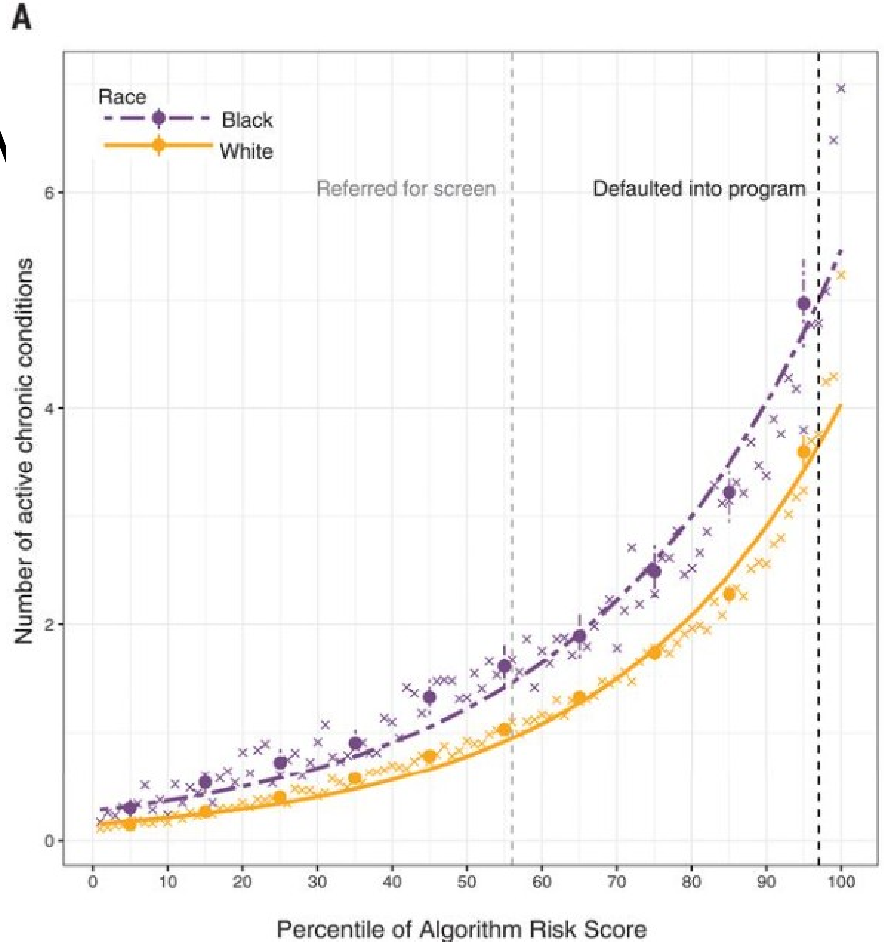
Case Study

- Common insurance provider
- Identify patients for “high risk care programs” via determining which patients will incur higher healthcare expenditures
- What fairness goals should we identify?
- What patient characteristics should be tracked and measured in the name of fairness?
- Should any patient data be disregarded and considered irrelevant to predictions?

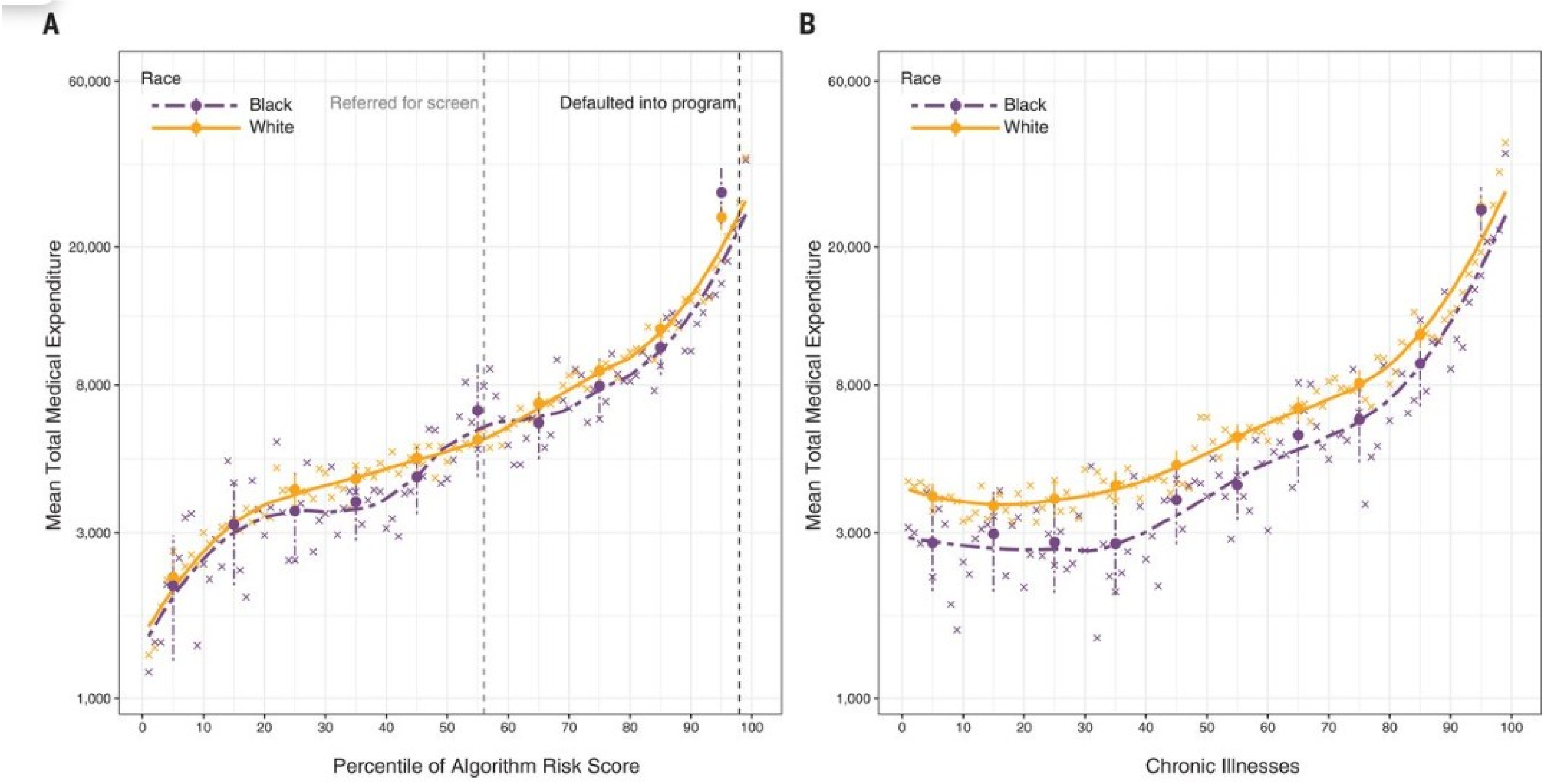
Case Study

- Primary care patients for 2 years from a major health system
- Assigned a given risk score based on PM





Case Study



A
Race
Black
White
Referred for screen
Defaulted into program
Mean Total Medical Expenditure
60,000
20,000
8,000
3,000
1,000
0
10
20
30
40
50
60
70
80
90
100
Percentile of Algorithm Risk Score
B
Race
Black
White
Mean Total Medical Expenditure
Chronic Illnesses

Results

- Did we choose a good result measure?
- Did this algorithm achieve its goal? Was it fair?
- Should the algorithm be blinded?
- Should we preprocess the data in any way?

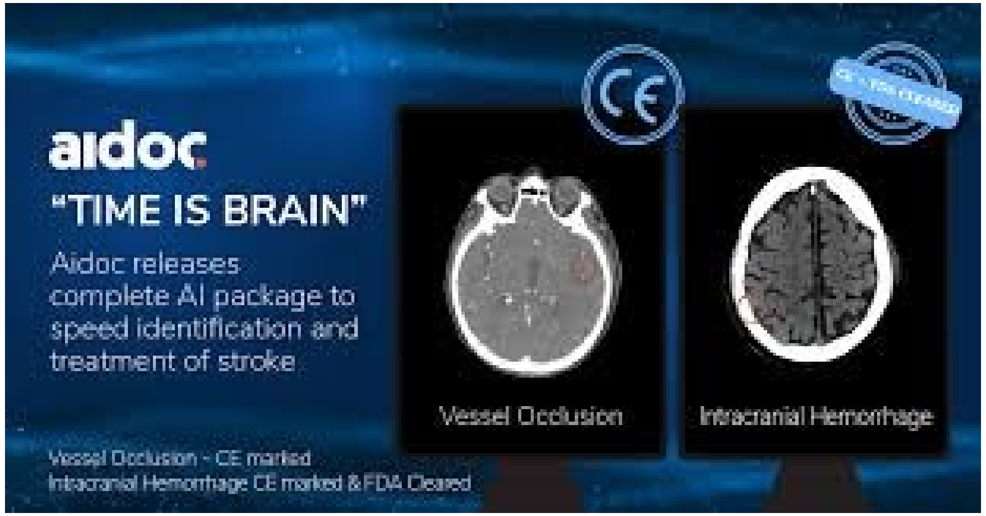
Pathways for Medical Device Approval

- 510k
- De Novo Approval
- Premarket Approval
- Exemptions (rare disease, breakthrough)

For the purposes of the FDA, software (including AI) is considered a medical device

510(k) Pathway for Medical Devices

- FDA approval process for medical devices
- Must find a “predicate” device with same use

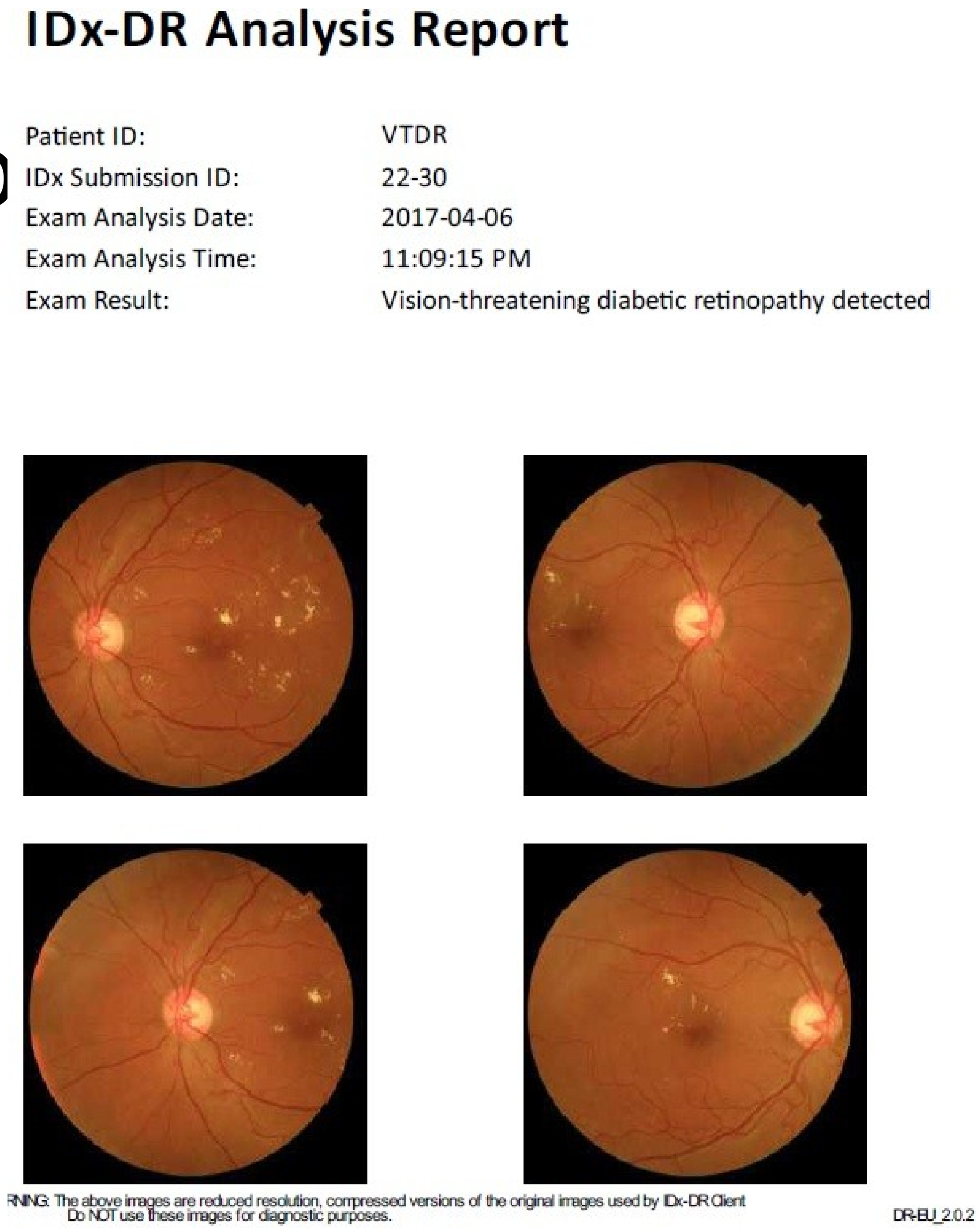


Addressing Data Drift

- Predetermined Change Control Plans (PCCPs)
- Plan outlining what changes and surveillance will occur ahead of time
- Not required for FDA approval

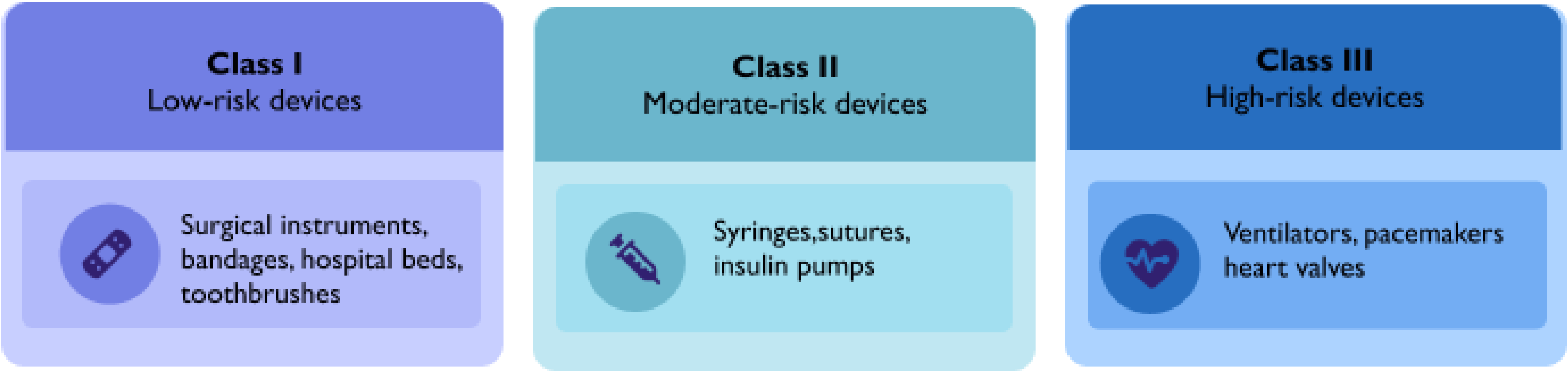
De Novo Device Appro

- For devices that have no precedent
- Ex. IDX: Diabetic retinopathy
- Once approved, can serve as precedent in future 510k



FDA Device Classification

- Class I: Low risk (e.g., stethoscope) – usually exempt.
- Class II: Moderate risk – majority of AI devices fall here.
- Class III: High risk – life-supporting or life-sustaining, needs PMA



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FDA Device Classification

- Class I: 510k eligible
- Class II: 510k eligible
- Class III: Needs pre-market approval

Pre Market Authorization

- Needed for high risk medical devices
- Design a study for use
- During the study, obtain an investigational device exemption for the experimental device
- Then, FDA reviews data for clinical approval

Next time:

- Project Intro
- Practice by coming up with a clinical question of your own

Thanks!